Card Club Arcade

From your computer to Google Play: the step-by-step guide

Version 17 · written for a first-time publisher · every step names the screen you are on and what to click

At a glance	
Total cost	**US$25, once**, for the Google Play developer account. Everything else in this guide is free: Node.js, Android Studio, GitHub Desktop, GitHub, Vercel, Firebase (later).
Your time	About 2 to 3 hours to reach the first upload. Then Google asks new personal accounts to run a **closed test with 12 testers for 14 days** (confirm the current rule in Play Console, it can change), followed by a review of a few days. Plan for 3 to 4 weeks from today to a public listing.
What I could not verify	The Android Gradle build itself (my workspace has no Android SDK), behaviour on a real phone (notch, full screen) and room codes across different networks. Part 4 tests exactly these on your phone.
Google rule you must meet	Since 31 August 2026 new apps must target Android 16 (API level 36). The project already does. Do not downgrade the Android tools.

Already done for you (inside card-club-arcade-v17.zip)	Only you can do (cannot be done for you)
15 games, bots, pass and play, room codes, hints, missions and coins, 321 automated tests passing	Create the Google Play developer account, pay US$25, verify your identity
Android app project in the folder android: Android 16 target, landscape only, full screen, real icon and splash	Install the free tools and build the signed app file on your PC (Part 4 and 5)
Privacy, terms and delete-data web pages, plus the Play Store pack in docs/play-store: listing text, Data safety and content rating answers	Choose the permanent app id and a support email, then run one command that fills them in
Store images in the folder store: icon, feature graphic, 7 phone screenshots	Test on a real phone and keep the signing key safe (two copies)
A cloud build workflow (GitHub Actions) as a backup way to build	Recruit 12 testers for the 14-day closed test and answer Google emails

TIP Words you will meet: **.aab** = the file you upload to Google Play. **Keystore (.jks)** = your signing key file; if you lose it you cannot update the app normally. **Package name / app id** = the permanent name of the app on Google Play (for example com.cardclubarcade.app). **Gradle** = the tool inside Android Studio that builds the app.

PART 1 Install the tools (once)

#	Where you are	What to do
1	**Browser: nodejs.org**	Click the green **LTS** button, run the installer, keep clicking Next, then Finish. Open Start, type **PowerShell**, press Enter, type `node -v` and press Enter. It must show **v22** or higher.
2	**Browser: developer.android.com/studio**	Click **Download Android Studio**, accept the terms, run the installer with the defaults. On first start: Import settings = **Do not import**, the wizard: choose **Standard**, Next, Finish. Wait while it downloads the Android SDK (several GB). You need a recent version (2025.2 "Otter" or newer).
3	**Browser: desktop.github.com**	Download **GitHub Desktop**, install, then File > Options > Accounts > **Sign in** with your GitHub account.
4	**Windows File Explorer**	Right-click **card-club-arcade-v17.zip** > Extract All. Choose a short folder such as **C:\CardClub** (long paths break Android builds).

PART 2 Prepare the project (about 10 minutes)

#	Where you are	What to do
1	**PowerShell**	Type `cd C:\CardClub` Enter, then `npm install` Enter. Wait 1 to 2 minutes.
2	**PowerShell**	Optional but wise: `npm test`. Expect **321 passed**.
3	**Decide: the app id**	The default **com.cardclubarcade.app** is fine. It is the permanent name on Google Play and must be unique worldwide. If you want your own (for example com.yourname.cardclub) change it **before the first upload** in three places: capacitor.config.ts (appId), android/app/build.gradle (namespace and applicationId) and the folder android/app/src/main/java/com/cardclubarcade/app plus the first line of MainActivity.java. If unsure, keep the default; if Play later says it is taken, ask me to change it.
4	**PowerShell**	Type `npm run legal -- "Your Full Legal Name" "yourname@gmail.com"` Enter. It prints "filled public/privacy.html" and so on. This puts your name and email into the privacy, terms and delete-data pages and the Play documents. Use an email you will check: it becomes public.
5	**Notepad**	Open **docs\play-store\LISTING.md** and replace **https://YOUR-DOMAIN** with your Vercel address (for example https://clubcardv2.vercel.app). Save.

PART 3 Put the website online (the privacy page needs a public link)

#	Where you are	What to do
1	**GitHub Desktop**	File > Clone repository > the **URL** tab > type **sakshamgolchha5/clubcardv2** (your repository) > choose a local path such as C:\Repos > Clone. Skip this if it is already cloned.
2	**File Explorer**	Copy everything from C:\CardClub into the cloned repository folder **except the folders node_modules and dist**. Click Replace the files when asked.
3	**GitHub Desktop**	The changed files appear on the left. At the bottom-left type the Summary **v17 Play Store pack**, click **Commit to main**, then click **Push origin** at the top.
4	**Browser: vercel.com > your project > Deployments**	A new deployment starts. Wait for the green **Ready** (1 to 2 minutes).

#	Where you are	What to do
5	**Browser**	Open **https://your-domain/privacy.html**, **/terms.html** and **/delete-data.html**. Your name and email must be there and no curly brackets {{ }} may remain. On your phone, open the site and check the game library loads.

NOTE Why GitHub Desktop and not "upload files" on the GitHub website: the website accepts only 100 files per upload and the Android folder has many more.

PART 4 Build and test the Android app in Android Studio

#	Where you are	What to do
1	**PowerShell, in C:\CardClub**	`npm run android:sync` builds the web app and copies it into the Android project. It ends with "Sync finished". Run it again every time the web app changes.
2	**PowerShell**	`npm run android:open` opens Android Studio on the android folder. If it asks **Trust and Open Project?** click **Trust Project**.
3	**Android Studio, bottom status bar**	Wait until the Gradle sync finishes (first time 5 to 15 minutes, it downloads files). If you see a Gradle JDK error: File > Settings > Build, Execution, Deployment > Build Tools > Gradle > **Gradle JDK** > choose the bundled **jbr-21** > Apply, then click the elephant icon **Sync Project with Gradle Files**. If a popup offers to update the Android Gradle plugin, choose **Remind me later**.
4	**Your phone**	Settings > About phone > tap **Build number** 7 times > go back > Developer options > turn on **USB debugging**. Plug the phone into the PC and tap **Allow** on the phone. No phone? In Android Studio open Device Manager (right side) > Create Virtual Device > Phone > Pixel 8 > Next > download a system image (API 35 or 36) > Finish.
5	**Android Studio top toolbar**	Open the device drop-down and pick your phone, then press the green **Run** triangle. The app installs and opens on the phone.
6	**Your phone: the test list**	Check: opens in landscape and full screen; the game library shows all 15 icons; play one game against bots; sound on and off in Settings; a hint (needs 200 coins); Missions tab; nothing hidden behind the camera notch; create a room code and join it from a second phone; press Home and reopen. **Send me anything odd (a screenshot is enough).**

PART 5 Make the signed release file (.aab)

#	Where you are	What to do
1	**Android Studio menu bar**	Build > **Generate Signed App Bundle / APK...**
2	**Dialog**	Choose **Android App Bundle** > Next.
3	**Key store path**	Click **Create new...**. Key store path: a folder **outside** the project, for example D:\Keys\cardclub-upload.jks. Fill Password (twice), Alias **upload**, Key password, Validity **25** years, and your first and last name > OK.
4	**Dialog**	Next > select **release** > **Create**. When the notification appears click **Locate**. The file is **android\app\release\app-release.aab**: this is what you upload to Google.
5	**File Explorer**	**Back up the .jks file now** to two places (for example a USB stick and a private cloud folder) and save the passwords in a password manager.

IMPORTANT Never commit the keystore to GitHub (the project already blocks *.jks) and never send it to anyone. Google Play App Signing is on by default: Google keeps the final app signing key and your file is only the upload key.

PART 6 Google Play Console: account, app and forms

#	Where you are	What to do
1	**Browser: play.google.com/console/signup**	Sign in with the Google account you want to keep for this. Choose **For personal use** (or Organization if you have a registered business). Enter developer name, contact email, phone and address. Google shows your developer name and email publicly.
2	**Payment page**	Pay the one-time **US$25** registration fee.
3	**Verification**	Upload the ID document and verify your phone number when asked. Approval can take hours to a few days; you will get an email.
4	**Console > All apps > Create app**	App name **Card Club Arcade: Card Games**. Default language English. **App or game: Game**. **Free or paid: Free**. Tick the Developer Program Policies and US export laws boxes > **Create app**. If Play says the package name is taken, tell me and I will change the app id.
5	**Left menu: Policy and programs > App content > Privacy policy**	Paste **https://your-domain/privacy.html** > Save.
6	**App content > App access**	Choose **All functionality is available without special access** > Save.
7	**App content > Ads**	Choose **No, my app does not contain ads** > Save.
8	**App content > Content rating**	Start questionnaire > your email > category **Game** > answer from **docs/play-store/CONTENT_RATING.md** (violence, sexual content, language, drugs: No; gambling and simulated gambling: No; users can interact: Yes; shares location: No; purchases: No) > Save > Submit > **Apply ratings**.
9	**App content > Target audience and content**	Tick **13-15, 16-17 and 18 and over** (not under 13) > Next > Appeals to children? No > Save. (Choosing under 13 brings the stricter Families rules.)
10	**App content > Data safety**	Start > follow **docs/play-store/DATA_SAFETY.md** line by line > Next > Preview > **Submit**. Read its note about room-code connections: if unsure, declare the connection details as collected for app functionality, not for ads.
11	**App content > other forms**	Government apps: No. Financial features: "My app does not provide any financial features". Health: none. News: not a news app. Advertising ID: the app does not use it.
12	**Left menu: Grow users > Store presence > Main store listing**	Paste the app name, short description and full description from **docs/play-store/LISTING.md**. Upload **store/icon-512.png** (App icon), **store/feature-graphic-1024x500.png** (Feature graphic) and the seven **store/phone-*.png** (Phone screenshots) > Save.
13	**Store presence > Store settings**	Category **Game > Card**, add tags, your contact email and website > Save.

PART 7 Closed test, then production

#	Where you are	What to do
1	**Left menu: Test and release > Testing > Closed testing**	Click **Create track** (name it Friends) > **Create new release**.
2	**Release page**	Under App bundles click **Upload** and choose app-release.aab. First time, accept **Play app signing** (Continue). Release notes: paste the text from LISTING.md "What's new". Click **Next**; red errors must be fixed, yellow warnings can wait; **Save**; then **Go to publishing overview** > **Send changes for review**.
3	**Closed testing > Testers tab**	Create an email list and add **at least 12** Gmail addresses of people who will really install it (friends, family, colleagues). Save. Copy the **Join on the web** link and send it. Each tester opens the link, taps **Become a tester**, then installs from the Play Store. They must stay opted in for **14 days in a row**.
4	**Dashboard**	When the test period is complete the **Apply for production** button appears. Answer the short questions (what you tested, what you changed after feedback) > Submit. Google replies by email.
5	**Test and release > Production > Create new release**	Click **Add from library** and pick your tested bundle (or upload a newer one) > release notes > Next > Save. Open the **Countries/regions** tab > Add countries (for example India and all others) > **Send for review**.
6	**Your inbox**	The first review can take several days. When the email says the app is live, open the Play Store link on your phone and check the listing.

TIP In the 14-day test, send me every bug or complaint your testers report. I fix them, you bump the version (Part 8) and upload to the same closed track. Fixing early is cheap; a rejected production review costs days.

PART 8 Releasing updates later

#	Where you are	What to do
1	**Notepad: android\app\build.gradle**	Increase **versionCode** (2, 3, 4 ...) and change **versionName** (for example 1.1). The number must be higher on every upload.
2	**PowerShell, Android Studio**	Run `npm run android:sync`, then Build > Generate Signed App Bundle > Android App Bundle > **Choose existing...** > pick your .jks (same key as before) > release > Create.
3	**Play Console > Production (or Closed testing) > Create new release**	Upload the new .aab > release notes > Next > Save > Send for review. Also push the same code to GitHub so the website matches.

PART 9 If something goes wrong

What you see	What to do
Gradle sync fails with a Java or JDK message	File > Settings > Build Tools > Gradle > Gradle JDK = jbr-21, then Sync Project with Gradle Files.
"SDK location not found"	File > Settings > Languages and Frameworks > Android SDK: note the path, create android\local.properties with the line sdk.dir=THAT_PATH (use double backslashes).

What you see	What to do
Your phone is not in the device list	Re-plug the cable, set USB mode to File transfer, accept the debugging prompt on the phone; or use an emulator.
The app opens but shows a blank screen	Run npm run android:sync again, then Run. Make sure npm run build finished without errors.
Play says "target API level too low"	Open android\variables.gradle: compileSdkVersion and targetSdkVersion must be 36 or higher. Do not downgrade Capacitor.
Play says the package name already exists	Choose another app id (Part 2, step 3) before the first successful upload, rebuild, and upload again.
Upload rejected: version code already used	Increase versionCode in android\app\build.gradle (Part 8).
A cloud (GitHub Actions) build fails	Open the failed run, copy the red log lines and send them to me; cloud Gradle builds usually need one fix.
You lost the keystore or its password	In Play Console open Test and release > App integrity and use "Request upload key reset". Keep backups to avoid this.

PART 10 Legal and policy checklist (practical, not legal advice)

Topic	What to do
Privacy policy must stay true	The pages in public/ say: no accounts, no ads, no analytics, data stays on the device, room codes use PeerJS and expose connection details to the other players. If you ever add accounts, cloud saves, ads, analytics or purchases, update the policy, the Data safety form and the content rating before releasing.
No real-money gambling	Teen Patti, Poker and betting features are excluded on purpose. Coins have no cash value and cannot be bought. If you ever sell coins or items for real money, Google requires Google Play Billing and a payments profile, and the rating and policy answers change.
Original assets	The icons, store images, sound effects (generated in code) and fonts (system fonts) in this project are original or system-provided. Do not add other companies' logos, names or screenshots (for example other games' brand names) to the listing or the app.
Player names	Display names are typed by users and shown to others. The terms page forbids offensive names. Later, add a report or mute option if the player base grows.
Children	Target audience is 13+. The app collects no personal data. Do not choose an under-13 audience without reading Google's Families policy.
India and EU privacy laws	Today you collect no personal data on a server, so obligations are light. If you add accounts or cloud sync you handle personal data: add a consent notice, a grievance contact and a delete-account option (India DPDP Act 2023; GDPR for EU users). Take advice then.
Developer account hygiene	Turn on two-step verification on Google, GitHub and Vercel. Keep the keystore backed up. Read Google Play policy emails promptly: missed deadlines can suspend the app.
Taxes and payments	Free app: nothing to set up. Before you earn anything (ads or purchases) you must complete a Google payments profile with tax details; ask your tax adviser how to report the income.

PART 11 Future plan

Phase	When	What happens	Cost
0. Launch	Now to week 4	You do Parts 1 to 7. In the closed test I fix what testers report, polish the real-phone feel (notch, speed, sounds) and add a Hindi listing.	US$25 once
1. More games	Weeks 4 to 8	4 or 5 new games from your Card Club list. Suggested order, cheapest first: Spades, Euchre, Gin Rummy, Tonk, Rummy 500; then Cribbage, Pinochle, Canasta. About 1 to 2 working steps each, tested and illustrated like the others. Also gems, a daily spin, more themes and card backs.	Free
2. Accounts and friends	Months 2 to 3	Real sign-in with Firebase Authentication (free Spark plan), unique usernames, cloud save of coins and progress, friend requests with online status, push-notification invites (Firebase Cloud Messaging), simple leaderboards. This needs a small backend, so privacy policy, Data safety and a delete-account feature must be updated first.	Free tier, then pay as you grow
3. Optional income	Month 4 onwards	Choose one: rewarded video ads (Google AdMob) or cosmetic purchases through Google Play Billing (never gambling, never cash value). Needs a payments profile, updated policy and Data safety, and the "contains ads" or "in-app purchases" labels on the listing.	AdMob and Billing are free to join; Google takes a share of purchases

PART A Your checklist and where the files are

- ☐ Install Node.js 22 LTS, Android Studio, GitHub Desktop (Part 1)
- ☐ Unzip v17 to C:\CardClub, run npm install (Part 2)
- ☐ Decide the app id (keep the default unless you have a reason)
- ☐ Run npm run legal with your legal name and a support email
- ☐ Replace YOUR-DOMAIN in docs/play-store/LISTING.md
- ☐ Commit and push to GitHub; wait for Vercel Ready; open /privacy.html /terms.html /delete-data.html (Part 3)
- ☐ npm run android:sync, open in Android Studio, run on your phone, finish the test list (Part 4)
- ☐ Build the signed .aab and back up the keystore twice (Part 5)
- ☐ Create the Google Play developer account, pay US$25, verify identity (Part 6)
- ☐ Create the app and fill every App content form, then the store listing with the images (Part 6)
- ☐ Upload the .aab to a closed test, add 12 testers, keep them opted in for 14 days (Part 7)
- ☐ Apply for production, release to your countries, share the Play link (Part 7)

File or folder in v17	What it is for
android/	The Android Studio project (open this folder in Android Studio).
capacitor.config.ts	App id, app name, bundled web folder.
public/privacy.html, terms.html, delete-data.html	The public legal pages Play needs (privacy URL, delete-data link).
docs/play-store/	LISTING.md (store text), DATA_SAFETY.md, CONTENT_RATING.md, RELEASE_CHECKLIST.md, ANDROID_BUILD.md.
store/	icon-512.png, feature-graphic-1024x500.png and seven 1920x1080 phone screenshots for Play Console.
.github/workflows/android.yml	Optional cloud build: GitHub > Actions > Android build > Run workflow.
scripts/fill-legal.mjs	The command behind npm run legal.
docs/STATUS.md	Full list of what is built, what is verified and what is not.

Labels in Google Play Console and Android Studio change from time to time; if a button is named slightly differently, look for the closest match and send me a screenshot if you are stuck.